C Programming

ITERATION, FOR LOOPS
The basic format of the for statement is,

```
for( start condition; continue condition; re-evaulation )
        program statement;
```

```
/* sample program using a for statement */
#include <stdio.h>

main()   /* Program introduces the for statement, counts to ten */
{
        int  count;

        for( count = 1; count <= 10; count = count + 1 )
                printf("%d ", count );

        printf("\n");
}
```

```
Sample Program Output
1 2 3 4 5 6 7 8 9 10
```

The program declares an integer variable *count*. The first part of the *for* statement

```
for( count = 1;
```

initialises the value of *count* to 1. The *for* loop continues whilst the condition

```
count <= 10;
```

evaluates as TRUE. As the variable *count* has just been initialised to 1, this condition is TRUE and so the program statement

```
        printf("%d ", count );
```

is executed, which prints the value of *count* to the screen, followed by a space character.

Next, the remaining statement of the *for* is executed

```
count = count + 1 );
```

which adds one to the current value of *count*. Control now passes back to the conditional test,

```
count <= 10;
```

which evaluates as true, so the program statement

```
        printf("%d ", count );
```

is executed. *Count* is incremented again, the condition re-evaluated etc, until count reaches a value of 11.

When this occurs, the conditional test

```
count <= 10;
```

evaluates as FALSE, and the *for* loop terminates, and program control passes to the statement

```
printf("\n");
```

which prints a newline, and then the program terminates, as there are no more statements left to execute.

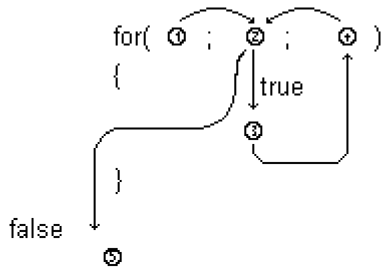
```
/* sample program using a for statement */
#include <stdio.h>

main()
{
        int  n, t_number;

        t_number = 0;
        for( n = 1; n <= 200; n = n + 1 )
                t_number = t_number + n;

        printf("The 200th triangular_number is %d\n", t_number);
}
```

Sample Program Output

```
The 200th triangular_number is 20100
```

The above program uses a *for* loop to calculate the sum of the numbers from 1 to 200 inclusive (said to be the *triangular number*).

The following diagram shows the order of processing each part of a *for*



An example of using a for loop to print out characters

```
#include <stdio.h>

main()
{
        char letter;
        for( letter = 'A'; letter <= 'E'; letter = letter + 1 ) {
                printf("%c ", letter);
        }
}
```

Sample Program Output

```
A B C D E
```

An example of using a for loop to count numbers, using two initialisations

```
#include <stdio.h>

main()
{
        int total, loop;
        for( total = 0, loop = 1; loop <= 10; loop = loop + 1 ){
                total = total + loop;
        }
        printf("Total = %d\n"
```